

Literature Benchmark Brief: Gaertner_2020_NREL

Title: Definition of the IEA Wind 15-Megawatt offshore reference wind turbine

Authors: E. Gaertner, J. Rinker, L. Sethuraman, F. Zahle, B. Anderson, G. E. Barter, K. Dykes (2020)

Venue: NREL Technical Report | **DOI:** 10.2172/1603478

Cluster: Cluster C: Aerodynamics & Control Cliff

Core Theoretical Mechanism & Model Formulation

IEA 15MW reference turbine specification, Region 2 to Region 3 transition parameters

Empirical Scale & Experimental Dataset

Global standard aeroelastic benchmark

Relevance to IEEE TSTE & Sustainable Energy Systems

Establishes authoritative industrial standards for rated wind speed and pitch regulation boundaries

Identified Gaps & Operational Vulnerabilities

Simulation specification without real-world SCADA noise and sensor degradation

Role in Manuscript Defense & Reviewer #2 Proof

Defends ground-truth operational regime partition logic in Section II-B